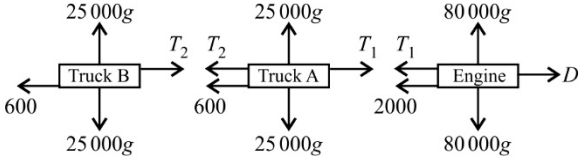
6	**(a)**			**B1** **B1** **B1** **[3]**	**1.1a** **1.1a** **1.1a**	Weights and normal reactions shown Tensions (must be two distinct) shown Driving force (D or 16 200) and three resistances correctly shown	Allow N_1, N_1 and N_2 instead of values to match weights for the normal reaction (condone N_1, N_2, N_3)
6	**(b)**		N2L for the whole train: $D - 2000 - 600 - 600 = 130\,000 \times 0.1$ $D = 16\,200$	 **M1** **A1** **[2]**	 **1.1a** **1.1**	All forces present and no extras– allow slip in the mass as long as it is their attempt at total mass of the train. Signs must be consistent	
			Alternative solution N2L for each part of the train $D - 2000 - T_1 = 80\,000 \times 0.1$ $T_1 - T_2 - 600 = 25\,000 \times 0.1$ $T_2 - 600 = 25\,000 \times 0.1$ Add equations to eliminate tension $D = 16\,200$	 **M1** **A1** **[2]**		All forces present and no extras in each equation Signs must be consistent Award the M mark only when an attempt is made to solve the equations simultaneously	Other equations may be used instead Both trucks A and B $T_1 - 2 \times 600 = 5000$ Engine plus A $D - T_2 - 2600 = 10500$
6	**(c)**		N2L for truck A: $T_1 - T_2 - 600 = 25\,000 \times 0.1$ Difference $T_1 - T_2$ is 3100 N	**M1** **A1** **[2]**	**3.1b** **1.1**	All forces present and no extras Signs must be consistent from correct working	Values for tensions need not be found
			Alternative solution For the engine $D - 2000 - T_1 = 8000$ $T_1 = 6200$ N For truck B $T_2 - 600 = 2500$ $T_2 = 3100$ N Difference $T_1 - T_2$ is 3100 N	**M1** **A1** **[2]**		All forces present and no extras in one equation leading to a value for either T_1 or T_2. Signs must be consistent FT their D	The equations may have been found in part (b) Also allow T_1 is twice T_2 oe from correct values